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# Comparison of the impact of dimensionality reduction and data splitting on classification performance in credit risk assessment

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**Abstract**
Credit risk assessment (CRA) plays an important role in credit decision-making process of financial institutions. Today, developing big data analysis and machine learning methods have marked a new era in credit risk estimation. In recent years, using machine learning methods in credit risk estimation has emerged as an alternative method for financial institutions. The past demographic and financial data of the person whose CRA will be performed is important for creating an automatic artificial intelligence credit score prediction model based on machine learning. It is also necessary to use features correctly to create accurate machine learning models. This article aims to investigate the effects of dimensionality reduction and data splitting steps on the performance of classification algorithms widely used in the literature. In our study, dimensionality reduction was performed using Principal Component Analysis (PCA). Random Forest (RF), Logistic Regression (LR), Decision Tree (DT), Naive Bayes (NB) algorithms were chosen for classification. Percentage splitting (PER, 66–34%) and k-fold (k = 10) cross-validation techniques were used when dividing the data set into training and test data. The results obtained were evaluated with accuracy, recall, F1 score, precision, and AUC metrics. German data set was used in this study. The effect of data splitting and dimension reduction on the classification of CRA systems was examined. The highest ACC in PER and CV data splitting was obtained with the RF algorithm. Using data splitting methods and PCA, the highest accuracy was observed with RF and the highest AUC with NB, with 13 PCs in which 80% of the variance was obtained. As a result, the data set consisting of a total of 20 features, expressed by 13 PCs, achieved similar or higher success than the results obtained from the original data set.

**Keywords** Principal component analysis · Data splitting with percentage and cross-validation · Credit risk assessment

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## 1 Introduction

Lending money is a traditional process. However, the creditor lends money based on several factors. By designing a credit scoring process, lenders can obtain information about an individual's repayment behavior. This process led to the development of systems in the 1960s that calculated the percentage risk associated with lending money, that is, the probability that customers would repay within a specified period. Credit assessment is used to estimate the possibility of individuals not being able to repay the borrowed amount by assessing the risk before lending money to individuals.

In recent years, banks or financial institutions that provide loans by meeting people's borrowing needs have been developing various credit scoring models to improve the assessment of an applicant's creditworthiness during the credit evaluation process. Credit scoring is a process by which creditors (banks, insurance companies) assign loan applicants to the "good" group, more likely to repay the debt, or to the "bad" group, more likely to default on the debt or any financial obligation, i.e. debt within a certain period of time.

With the growth of the credit sector, the importance of different risk assessment models has increased in the sector. It is very important that these models produce accurate, sensitive and robust results. For this reason, predictive machine learning models have become a very popular research topic in the field of credit in recent years. Today, the use of machine learning techniques to create credit scoring models is rapidly increasing. Using demographics, historical data on payments, and statistical techniques, these models can help identify key demographic characteristics related to credit risk and assign a score to each customer. The probability of an applicant defaulting must be calculated based on the information provided at the time of application by the applicant, forming the foundations of the applicant's creditworthiness.

When the existing studies in the literature are examined, it is seen that PER or CV are preferred among the data splitting methods. It has been observed that using these two methods together and evaluating the results is a subject open to research. Another motivation is to examine and evaluate the use of dimensionality reduction together with data splitting methods. In studies carried out with this motivation, different results were obtained specifically for the classification algorithm.

In this study, the effect of using data splitting (PER and CV) and dimension reduction (PCA) in the classification of CRA systems was examined. Using the German credit data set, classification was made with LR, DT, RF and NB algorithms. The results obtained from ACC, PER, REC, F1 Score and AUC evaluation metrics were examined.

The remainder of the paper is organized as follows. Section 2 of this article includes a literature review on credit risk assessment with machine learning methods. Section 3 includes Materials and Methods, Sect. 4 contains Experimental Results. Finally, the last section concludes our paper and states final remarks.

## 2 Literature review

This section examines studies conducted for credit risk assessment (CRA). In these studies, many data sets and machine learning methods were used to evaluate these data sets. The data sets included in this study are summarized in Table 1 in terms of their content.

**Table 1** Studies on CRA with machine learning methods

| References | Dataset type | Classification model | Evaluation metrics | ACC result (%) |
|---|---|---|---|---|
| Appel et al. (2019) | Invoice | NB, LR, k-NN, RF, GBDT | AUC–ROC, PRC, CM | 77 |
| Ampountolas et al. (2021) | Credit | AdaBoost, XGBoost, DT, ETC, RF, k-NN, NN, MLP | PREC, RD , F1, feature importance, hyper-parameter optimization | 81 |
| Moscato et al. (2021) | P2P | LR, RF, MLP | G-Mean, AUC, REC, SPEC, ACC | N/A |
| Hindistan et al. (2019) | P2P | LR, RF, DT | ACC, PREC, REC, F1 | 97 |
| Wu and Pan (2021) | P2P | LR, RF, SVM | CM, PREC, RECl, AUC, K-S value | 75 |
| Kruppa et al. (2013) | Credit | RF-PET, LR, Runned LR, k-NN, Bagged k-NN | ROC curves, AUC, BS | 95 |
| Pławiak et al. (2020) | Credit | DGHNL, SVM, k-NN, Probabilistic NN, Fuzzy System (FS) | CM, ACC, AUC, ROC | 95 |
| Bao et al. (2019) | P2P and Credit | LR, DT, RF, GBDT, SVM, k-NN, Artificial NN, k-means, Kohonen's SOM | ROC, MCC, AUC, ACC, REC, PREC, Type I and II errors | N/A |
| Zhang et al. (2019) | Credit | LDA, LR, SVM, DT, MLP, EMPNGA | ACC, AUC, H-measure, BS | N/A |
| Abellán and Castellano (2017) | Credit | LR, MLP, SVM, DT, Credit DT, AdaBoost, Bagging, RSP, DECORATE, Rotation Forest (RotF) | AUC | N/A |
| Xia et al. (2020) | Credit | LR, ANN, DT, SVM, Bagging-ANN, RF, GBDT, XGBoost, LightGBM, CatBoost | ACC, AUC, H-measure, BS | N/A |
| García et al. (2019) | Credit | Bagging, AdaBoost, SGBoost, RSP, RotF, RF, DECORATE | TP, TN, FP, FN, Type I and II errors, AUC, ACC | N/A |
| Zhao et al. (2015) | Credit | MLP | ACC | 87 |
| Luo et al. (2017) | Credit | Multinomial LR, MLP, SVM, DBN with RBM | AUC, ACC | 100 |
| Boughaci and Alkhawaldeh (2020) | Credit | k-NN, RF, BNC, NB, MLP, LR, SVM, OneR, Bagging, AdaBoost, LogitBoost | ROC, PCR, TPR , FPR , PREC, REC, F-measure, MCC | 94 |
| Niu et al. (2019) | Social Network and P2P | LR, RF, AdaBoost, LightGBM | ACC, AUC, F1 | 71 |
| Zhu et al. (2019) | P2P | RF, DT, SVM, LR | ACC, REC, F1, ROC, AUC | 98 |
| Djeundje et al. (2021) | Demography and E-mail | LR, LASSO, XGB, NN | AUC | N/A |

**Table 1** (continued)

| References | Dataset type | Classification model | Evaluation metrics | ACC result (%) |
|---|---|---|---|---|
| Liu et al. (2021) | Credit | AdaBoost, DT, LR, NN, SVM, RF, GBDT, mg-GBDT | ACC, AUC, BS, PREC, REC | 77 |

*NB* Naive Bayes, *LR* Logistic Regression, *k-NN* k-Nearest Neighbors, *RF* Random Forest, *GBDT* Gradient Boosted Decision Trees, *AUC–ROC* Area Under the Receiver Operating Characteristic Curve, *PRC* Precision–Recall Curve, *CM* Confusion Matrix, *PREC* Precision, *REC* Recall, *SPEC* Specificity, *DT* Decision Tree, *ETC* Extra Trees Classifier, *NN* Neural Network, *MLP* Multi-Layer Perceptron, *SVM* Support Vector Machine, *ACC* Accuracy, *RF-PET* RF Probability Estimation Trees, *BS* Brier Scores, *DGHNL* Deep Genetic Hierarchical Network of Learners, *SOM* Self-Organizing Maps, *MCC* Matthews Correlation Coefficient, *EMPNGA* Enhanced Multi-Population Niche Genetic Algorithm, *DECORATE* Diverse Ensemble Creation by Oppositional Relabeling of Artificial Training Examples, *SGBoost* Stochastic Gradient Boosting, *RSP* Random Subspace, *BNC* Bayes Network Classifier, *DBN* Deep Belief Network, *RBM* Restricted Boltzmann Machines

Appel et al. (2019), presented a solution prototype developed during a partnership with a multinational bank to predict bill payments from collectors. A data set consisting of 175.552 invoices and 3.725 customers from 8 countries in Latin America was used. Tests were carried out with NB, LR, k-NN, RF and GBDT from among the supervised learning algorithms, and it was observed that the best models were RF and GBDT with an average accuracy of 77%.

Ampountolas et al. (2021), worked with various machine learning algorithms on real microcredit data to investigate their effectiveness in classifying borrowers into various loan categories. It has been shown that off-the-shelf multi-class classifiers such as RF can provide good results based on data about customers (such as age, occupation, and location). In the present study which was conducted using a data set of 4.450 customers of a microcredit company in Ghana, Correlation-based feature selection (CFS) and AdaBoost, XGBoost, DT, ETC, RF, k-NN, NN and MLP classification methods were used. Testing was performed with 5-fold cross-validation. It was found that XGBoost, AdaBoost and RF classifiers achieved almost the same prediction accuracy (80% on average).

Moscato et al. (2021), worked with data from a social lending platform, also known as Peer to Peer (P2P), consisting of 877.956 samples. The most used credit risk scoring models to predict whether a loan will be repaid on a P2P platform are compared. The study focused on the class imbalance problem and various resampling methods such as [no-resampling, under-sampling (RUS, IHT), over-sampling (ROS, SMOTE, ADASYN), hybrid-method (SMOTE-TOKEN, SMOTE-EN)]. Also, the study employed LR, RF and MLP classification methods. Classification results were interpreted with Explainable Artificial Intelligence (XAI) tools (LIME, ANCHORS, SHAP, BEEF, LORE). The best results were obtained with RUS and RF with 71% accuracy. XAI tools yielded different accuracies in different combinations.

Hindistan et al. (2019), investigated the use of machine learning techniques on big data platforms by analyzing credit scoring methods. A data set consisting of over 2 million data and 145 attributes, covering the years 2007–2015, was used from a Lending Club. It was concluded that in a Hadoop Distributed File System (HDFS) environment, LR outperformed DT and RF by 97% for credit scoring and classification, considering performance metrics such as accuracy, precision and recall, and the overall running time of the algorithms. LR performed better in terms of time in a single-node HDFS configuration compared to a non-HDFS configuration.

Wu and Pan (2021), worked on a data set consisting of 96.781 samples and 145 attributes belonging to a lending club, and focused on 5-fold cross-validation and RF, SVM and LR methods. It was observed that LR gives the best performance with 75%.

Kruppa et al. (2013), used the data set of a company that produces home appliances and offers its customers the opportunity to pay in installments. Due to the increasing receivable losses in installment loans, it is aimed to find a method to minimize this risk. The data set consists of 64.524 installment purchases of home appliances, each repaid in 13, 14, or 15 consecutive monthly installments. In this study where LR, RF-PET, k-NN algorithms were used, the most successful result was obtained with the RF-PET algorithm with 95% accuracy.

Pławiak et al. (2020), proposed DGHNL, a new deep learning model for credit risk prediction. This model aimed to provide a better performance by combining deep learning methods and genetic algorithms. The DGHNL model consists of two different stages. The first stage makes the data suitable for the model by preprocessing and feature engineering. The second stage determines the best feature set and model parameters using genetic algorithms and finally trains the deep learning model. In this article, German credit data

set was used. The performance of the DGHNL model was demonstrated by comparing it with other popular credit score prediction models, and the results showed that the DGHNL model provides higher accuracy, recall, and specificity. Testing was done with 10-fold cross-validation. As a result, it was concluded that the DGHNL model stands out as an effective deep learning model that can be used in financial predictions such as credit score prediction.

Bao et al. (2019), proposed a combination strategy to integrate unsupervised learning and supervised learning for credit risk assessment. Three different datasets were used. These datasets included German (1.000 samples and 20 attributes), Australian credit (690 samples and 14 attributes), and Chinese P2P (23.435 samples and 81 attributes). The study used supervised learning methods including LR, DT, RF, GBDT, SVM, k-NN, and ANN and unsupervised learning methods including k-means and SOM. The study was tested with 5-fold cross-validation. It was observed that using supervised and unsupervised learning methods together gives positive results for CRA.

Zhang et al. (2019), proposed a new multi-stage hybrid model that combines feature and classifier selection to obtain the optimal feature subset and optimal classifier subset, and then uses the classifier ensemble to improve the prediction performance based on these two optimal subsets. An enhanced multi-population niche genetic algorithm (EMPNGA) was proposed to effectively enhance the optimization ability by improving the selection, crossover and mutation steps and adding niche and transition steps. Moreover, EMPNGA was applied to combine various filter methods and prior information in feature and classifier selection to further improve the search efficiency. The proposed model was applied to Australian, German, Japanese, PPDai and GMSC credit datasets with 10-fold cross-validation to verify the prediction performance. Recommended model; LDA, LR, SVM, DT and MLP were compared with base classifiers. Experimental results confirmed that the performance of the proposed model is superior to other comparative models, arguing that this study is meaningful and effective.

Abellán and Castellano (2017), tried ensembles of classifiers and base classifiers together. Australia, Germany, Iran, Japan, Poland, and UCSD data sets were used. The main classifiers considered in this experimental included LR, MLP, SVM, C4.5, and CDT. The analyzed ensemble algorithms included AdaBoost, Bagging, RSP, DECORATE, and RotF. CDT showed the best performance with 5-fold cross-validation.

Xia et al. (2020), proposed a new tree-based dynamic heterogeneous ensemble method for credit scoring. The basic learning algorithms used in this study are C4.5, CART, CHAID, RF, and GBDT. This method aims to achieve a higher classification performance by combining multiple learning algorithms with different features. It is also specifically designed to deal with dynamic data changes, such as adding new data over time. The proposed method first generates small subsample data using a set of basic learning algorithms. Many different classifiers are then created for each subsampling using a tree-based learning algorithm. These classifiers are then combined to form a heterogeneous ensemble. Then, when new data is added, the proposed method is dynamically retrained. This process is accomplished by using only a portion of the new data, which increases the performance of the model while reducing the computational cost. The paper tested the performance of the proposed method on various real-world datasets. The results show that the proposed method achieves a higher classification performance than other similar techniques. Australia, Japan, Germany, PPDai, GMSC data sets were used. 5-Fold cross validation was carried out in the study. Basic algorithms including LR, ANN, DT, SVM and homogeneous ensemble algorithms including Bagging-ANN, RF, GBDT, XGBoost, LightGBM, CatBoost were used.

García et al. (2019), investigated the effect of sample types to improve the performance of ensemble models for credit risk and company bankruptcy prediction. The algorithms used in the article are basically decision trees and ensemble models. Decision trees are an algorithm that performs classification or regression operations using a tree structure that divides data into separate branches. Ensemble models are created by combining multiple models. Combining these models provides higher performance than a single model. The present study investigated performance differences between ensemble models created using different sampling techniques (random sampling, undersampling and upsampling). The study showed that different sampling techniques can improve the performance of ensemble models due to their different properties. In particular, it was found that undersampling techniques can improve performance, random sampling techniques can reduce variability, and upsampling techniques can cause overlearning. The study also used two different data sets to investigate the impact of sampling techniques. These datasets are the data set for business bankruptcy prediction and the data set for commercial credit risk prediction. The experiments conducted on two different data sets showed that subsampling methods further increase the performance of ensemble models. In conclusion, the study showed that it can be useful to use different sampling techniques to improve the performance of ensemble models. However, the study presented that the sampling technique that will be used may vary depending on the characteristics and purpose of the data set.

Zhao et al. (2015), conducted a study on automatic credit score prediction with MLP. German credit data set was used in the study. It focused on improving credit scoring models in three aspects. Firstly, it optimized data distribution in datasets using a new method called Average Random Choosing. Then, the effects of training-validation-test sample numbers were compared and it was aimed to find the most appropriate number of hidden layers. As a result, it has been determined that the optimization of the dataset and the increase in the number of hidden layers have a positive contribution to the study.

Luo et al. (2017), introduced a deep learning approach to credit scoring. A machine learning model was created using credit default swap (CDS) data (belonging to 661 companies from 6 regions). The deep learning model used in the article is an artificial neural network consisting of multilayer neural networks. CDSs are derivative products used to measure the bankruptcy risk of a company. This model was found to provide better results than traditional methods for credit scoring. In the article, the importance of CDS data was emphasized and it was stated that these data are less noisy and provide better predictions than traditional credit scoring data. Building the model using a deep learning approach enabled it to capture more complex relationships. In conclusion, this article showed that the use of CDS data for credit scoring is possible with a deep learning model that provides more accurate results. It was argued that this method may be a better tool for credit risk management.

Boughaci and Alkhawaldeh (2020), worked on credit scoring using machine learning methods. Tests were conducted on Germany, Australia, Japan, Poland, India, Taiwan, Give Me Some Credit Datasets (GMSCD) datasets with k-NN, RF, BNC, NB, MLP, LR, SVM, OneR, Bagging, AdaBoost, and LogitBoost algorithms. As a result of the numerical analysis, it was determined that there was no method which can consistently perform better than others in all data sets. Considering all the results, it was concluded that Bayes Net and Boosting classifiers are generally effective models for credit scoring.

Niu et al. (2019), used LR to test the relationship between borrowers' social network information and credit status. RF, AdaBoost and LightGBM were used to demonstrate the prediction performance of social network information. Examining the LR results shows that there is a statistically significant relationship between social network information and

loan default. Machine learning algorithm results showed that social network information can significantly improve loan default prediction performance.

Zhu et al. (2019), used RF, LR, DT and SVM algorithms. The Lending Club data set from the first quarter of 2019 was selected for the study. The experiment shows that the RF algorithm performs better than the other three algorithms in the prediction of loan default and has a strong ability of generalization. 98% ACC and AUC were observed with the RF algorithm.

Djeundje et al. (2021), evaluated the predictive accuracy of models using alternative data, that may be used instead of credit history, to predict the credit risk of a new account. This paper showed that a model that contains email usage and psychometric variables, as well as demographic variables, can give greater predictive accuracy than a model that uses demographic data only and that the predictive accuracy is sufficiently high for the demographic and email data to be used. They used LR, LASSO, XGB and NN algorithms.

Liu et al. (2021), used Australian, German, Japanese, Taiwan, Lending Club and WE datasets and proposed a step-wise multi-grained augmented gradient boosting decision trees (mg-GBDT) method. Results demonstrate that mg-GBDT can be a good choice for accurate credit scoring interpretation. ACC, AUC, BS, PREC, REC were used for evaluation metrics.

In the literature review, it was observed that cross validation or percentage were used as data splitting methods in CRA. We decided that observing the performance of CV and PER methods together is a subject open to research. Another motivation for our work includes dimension reduction with PCA. Performance analysis was performed with PCs with cumulative variance of 80% and more and positive results were observed. Our study has important results in terms of evaluating CRA transactions using data splitting and dimension reduction.

## 3 Materials and methods

The present study was carried out in the Python environment. All calculations were performed on a computer system with macOS 13.4 operating system, 16 GB RAM and Intel CORE i7 processor.

In our study, the following operations were performed respectively:

(1) Classification of data using PER,
(2) Classification of data using CV,
(3) Classification of data using PCA and PER,
(4) Classification of data using PCA and CV.

The procedures performed in this study are summarized in Fig. 1. The data set was first divided by PER and CV and evaluated with ACC, PREC, REC, F1 Score and AUC. Then, dimension reduction was performed with PCA and the results obtained with the same split methods and evaluation metrics were evaluated.

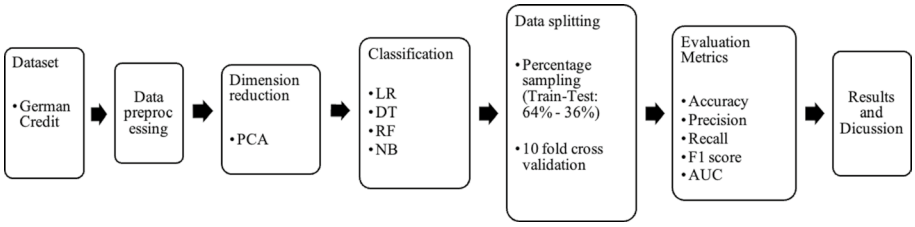


**Fig. 1** CRA model scope

**Table 2** Dataset detail

| Attribute name | Type | Data type | Description |
|---|---|---|---|
| checking_status | Categorical | String | Current account status |
| Duration | Discrete | Number | Credit period |
| credit_history | Categorical | String | Credit history |
| Purpose | Categorical | String | Credit purpose |
| credit_amount | Discrete | Number | Credit amount |
| savings_status | Categorical | String | Savings account/bonds |
| Employment | Categorical | String | Current business status |
| installment_commitment | Discrete | Number | Disposable income/installment ratio |
| personal_status | Categorical | String | Marital status and gender |
| other_{p}arties | Categorical | String | Other debts/guarantors |
| residence_since | Discrete | Number | Residence |
| property_magnitude | Categorical | String | Property ownership status |
| Age | Discrete | Number | Age |
| other_{p}ayment_{p}lans | Categorical | String | Other installment plans |
| Housing | Categorical | String | Whether or not you own a house |
| existing_credits | Discrete | Number | Number of credits available |
| Job | Categorical | String | Whether or not you have a job |
| num_dependents | Discrete | Number | Number of dependents |
| own_telephone | Categorical | String | Whether or not you have a phone |
| foreign_worker | Categorical | String | Whether there are foreign employees |
| Class | Categorical | String | Whether or not it is suitable for credit |

## 3.1 Dataset

Within the scope of this study, the publicly available German Credit Scoring dataset, which is one of the most used datasets in the literature, was selected (Hofmann 1994). The details of the dataset are shown in Table 2. The data set includes 1000 samples and 21 attributes (20 input/features and 1 output/target) that have previously received credit. The target variable distinguishes those who pay their loan debt on time and those who do not. 700 Samples consist of people who pay on time, and 300 samples consist of people who do not pay on time. Machine learning classifiers with feature selection techniques were trained on 66% of the data

set. 34% Data was used to test the classifiers and 10-fold cross-validation was used to evaluate the models.

### 3.2 Data preprocessing

Data preprocessing is a stage in machine learning processes that includes cleaning, filtering, integration, discretization and normalization of the data set in order to improve the quality of the data set and obtain more meaningful results (García et al. 2016). In this study, categorical attributes were converted to numerical values only for PCA as a pre-processing step.

### 3.3 Dimension reduction

High-dimensional datasets present some challenges in terms of computation, design, and interaction. One of the frequently used methods to solve these difficulties is to reduce a high-dimensional dataset to a low-dimensional space while preserving the properties of the high-dimensional structure. Such dimensionality reduction algorithms are useful abstractions because some dimensions in the dataset may not be necessary to understand the underlying patterns in the dataset. Instead, a subset of dimensions can be selected or learned to identify important features of the dataset (Fodor 2002).

#### 3.3.1 Principal component analysis (PCA)

PCA aims to linearly project the $N \times d$ input matrix $X = [X_1, \ldots, X_n]^T$ through an orthogonal transformation. The first dimension reflected defines the highest variance of the data, while the second dimension of the predicted data defines the second highest variance, and the analysis continues in this way. The first vector in the orthogonal transformation matrix is the first eigenvector representing the variance explained by the 1st principal component. Therefore, PCA can be used to reduce the dimensionality of data by removing the dimensions that describe the lowest variance in the data and retaining the dimensions that describe the highest variance in the data (Wold et al. 1987). The 1st eigenvector is calculated as follows (1):

$$V_1 = \underset{||V_1|| = 1}{Maximize}\left( \frac{V_1^T\,(X - mean(X))^T\,(X - mean(X))V_1}{V_1^T\,V_1} \right). \tag{1}$$

According to Eq. (1); $V_1$ is maximized when $V_1$ is the first eigenvector of data $X_1$. The subsequent eigenvectors $V_j$ are calculated as follows (2):

$$\begin{aligned} \hat{X}_{j-1}^T &= X - \sum_{i=1}^{j-1} X V_{j-1} V_{j-1}^T, \\ V_j &= \underset{||V_1|| = 1}{Maximize}\left( \frac{V_j^T\,\hat{X}_{j-1}^T\,\hat{X}_{j-1} V_j}{V_j^T\,V_j} \right). \end{aligned} \tag{2}$$

Before performing PCA, the input data matrix $X$ must be standardized to have a zero mean. The predicted lower dimensional representation of the input data $X$ is created by multiplying the input data $X$ by the selected eigenvectors $V_j$:

$$X_{proj} = XV_j. \tag{3}$$

### 3.4 Classification

In this section, the models that will be used in this study, which are among the most used machine learning methods in the literature, are explained to determine the best model for CRA.

#### 3.4.1 Decision tree (DT)

Decision tree is a classification model of computation based on entropy function and information gain. Entropy calculates the amount of uncertainty in data (4). Here $D$ is the current data and $q$ is a binary label from 0 to 1, and $p(x)$ is the ratio of the label $q$. To measure the entropy difference from the data, we calculate the information gain as shown in Eq. (5). Here $v$ is a subset of the data (Tu and Chung 1992).

$$E(D) = \sum_{i=1}^{m} - p\left(q_i\right) \cdot \log\left(p\left(q_i\right)\right), \tag{4}$$

$$I = E(D) - \sum_{v \in D} p(v)E(v). \tag{5}$$

#### 3.4.2 Random forest (RF)

RF is considered an advanced technique of DTs. The main idea of RF is to combine bagging and random subspace feature selection to combine individual DTs. The RF model uses randomness in two stages. Firstly, it randomly selects subsets from the original dataset. Secondly, feature subsets derived from the original full feature dimensions are randomly selected. Thus, the correlation between DTs in the forest is reduced. The final decision is made according to the voting procedure, where the input sample will be labeled as the class that received the majority of votes (Breiman 2001).

#### 3.4.3 Logistic regression (LR)

Logistic regression is a simple parametric statistical approach and is considered the industry standard in the field of credit scoring. It is used to solve binary classification problems (default and non-default in this work) and regression problems. The goal of the LR model is to obtain the logarithm of the ratio of two probability outcomes of interest (Lessmann et al. 2015).

The LR model for the independent variable p can be written as (6).

$$P(Y = 1) = \frac{1}{1 + e^{(-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_p X_p))}}. \tag{6}$$

Here $P(Y = 1)$ is the probability and $\beta_0, \beta_1, \ldots, \beta_p$ are the regression coefficients. There is a linear model hidden within the logistic regression model. The natural logarithm of the ratio of $P(Y = 1)$ to $1 - P(Y = 1)$ t gives a linear model in $X_i$ (7):

$$g(x) = \ln\left(\frac{P(Y=1)}{1-P(Y=1)}\right) - \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p. \tag{7}$$

$g(x)$ Has many of the desirable properties of a linear regression model. Independent variables can be a combination of continuous and categorical variables (Hosmer et al. 2013).

#### 3.4.4 Naive Bayes (NB)

It is a simple probabilistic classifier based on Bayes theorem, with assumptions of independence between features or variables. To derive a conditional probability for the relationships between feature values and class, NB analyzes the relationship between each feature and class for each instance (Rennie et al. 2003).

Let $x = (x^1, x^2, \ldots, x^d)$ be a d-dimensional sample without a class label, and our goal is to create a classifier that will predict the unknown class label based on Bayes theorem. Let $C = \{C_1, C_2, \ldots, C_k\}$ be the set of class labels. $P(C_k)$ is the prior probability of $C_k$ $(k = 1, 2, \ldots, k)$ inferred before new evidence. Let $P(x|C_k)$ be the conditional probability of seeing evidence x if hypothesis Ck is true. It is a technique for creating such classifiers using Bayes' theorem to obtain (8).

$$P\left(C^k \middle| x\right) = \frac{P\left(x|\ C_k\right)P(C_k))}{\sum_{k'} P(x|C_{k'})P(C_{K'})}. \tag{8}$$

NB assumes that the value of a particular property of a class is not related to the value of any other property (9).

$$P\left(x \middle|\ C_k\right) = \prod_{j=1}^{d} P(x^j|C_k). \tag{9}$$

### 3.5 Data splitting

One of the most important factors that increase classification success is the separation of data into training and test data. The literature suggests to split the data samples into 66% and 34% for training and testing, respectively (Picard and Berk 1990). In this study, the data was split between 66 and 34% for training and testing.

The second data splitting method used in the study is 10-fold cross validation. In 10-fold cross-validation, all data is divided into 10 parts by random sampling. First, classifiers are trained with 10% data and 90% data is taken to test the models. This process is repeated for 20–80, 30–70,...,70–30, 80–020 and 90–10 for the training set and test set, respectively. Finally, the results are evaluated for each environment and combined with the average for the final result (Fushiki 2011).

### 3.6 Evaluation metrics

Evaluation metrics are quantitative measurements used to evaluate the performance and effectiveness of a statistical or machine learning model. These metrics provide information about how well the model is performing and help compare different models or algorithms.

The confusion matrix is an $N \times N$ matrix, where $N$ is the number of predicted classes. For the relevant problem, we have $N = 2$ and hence we get a $2 \times 2$ matrix. It is a performance measure for machine learning classification problems where the output might be two or more classes. The confusion matrix is a table containing four different combinations of predicted and actual values. Extremely useful for measuring Recall, Specificity, Accuracy, and most importantly AUC–ROC curves (Sammut and Webb 2011).

Some of the metrics that can be obtained from the confusion matrix (Table 3) and that we used in this study are described below. True Positive (TP) refers to the number of correctly predicted positives. False Positive (FP) refers to the number of incorrectly predicted positives. True Negative (TN) refers to the number of correctly predicted negatives. False Negative (FN) refers to the number of incorrectly predicted negatives.

**Accuracy (ACC):** it is the ratio of the total number of correct samples to the number of correctly predicted samples. ACC measures the overall predictive effectiveness of models. However, it is not a reliable parameter as it can give misleading results if the dataset is not balanced, which is often seen in real-world credit datasets.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}. \tag{10}$$

**Precision:** it is the parameter that indicates success in a positively predicted situation.

$$Precision = \frac{TP}{TP + FP}. \tag{11}$$

**Recall:** it is the parameter that shows how successfully positive situations are predicted.

$$Recall = \frac{TP}{TP + FN}. \tag{12}$$

**F1 score:** it is the harmonic mean of precision and recall values for a classification problem.

$$F1\,Score = 2 * \frac{precision * recall}{precision + recall}. \tag{13}$$

**The area under the curve (AUC):** a receiver operating characteristics (ROC) chart is a technique for visualizing, organizing, and selecting classifiers based on their performance. The ROC curve is a two-dimensional representation of classifier performance. It is an efficient method for comparing classifiers to reduce ROC performance to a single scalar value representing expected performance. AUC is obtained by calculating the area under the ROC curve. Since AUC is a fraction of the area of the unit square, its value will always be between 0 and 1 (Fawcett 2006).

**Table 3** Confusion matrix

| True label | Predicted label | |
|---|---|---|
| | Positive | Negative |
| Positive | True positive (TP) | False negative (FN) |
| Negative | False positive (FP) | True negative (TN) |

**Table 4** Results obtained with PER

| | DT | LR | RF | NB |
|---|---|---|---|---|
| Accuracy (%) | 0.67 | 0.70 | 0.74 | 0.67 |
| Precision (%) | 0.78 | 0.89 | 0.92 | 0.78 |
| Recall (%) | 0.76 | 0.73 | 0.76 | 0.76 |
| F1 score (%) | 0.77 | 0.80 | 0.83 | 0.77 |

**Table 5** Results obtained with 10-fold CV

| | DT | LR | RF | NB |
|---|---|---|---|---|
| Accuracy (%) | 0.71 | 0.71 | 0.75 | 0.71 |
| Precision (%) | 0.80 | 0.76 | 0.77 | 0.81 |
| Recall (%) | 0.80 | 0.88 | 0.91 | 0.77 |
| F1 score (%) | 0.79 | 0.81 | 0.84 | 0.79 |

## 4 Experimental results

The German credit data set was first classified with DT, RF, LR and NB algorithms. PER (66–34%) and 10-fold cross-validation methods were used for data splitting. The results obtained were evaluated with accuracy, recall, F1 score, precision, and AUC metrics. Then, dimensionality reduction was performed with PCA and the same steps were repeated.

**Observation 1:** Classification with percentage sampling and 10-fold cross-validation

Tables 4 and 5 show the results obtained using percentage sampling and 10-fold cross-validation. In percentage sampling, in all evaluation metrics RF algorithm had the highest rates with 74% accuracy, 92% precision, 76% recall, and 83% F1 score. Additionally, when all metrics were examined, the second highest rates were obtained in the LR algorithm. DT and NB algorithms gave the same results in percentage sampling. When 10-fold cross validation was applied, the highest accuracy, recall, and F1 score of 75%, 91%, and 84%, respectively were obtained in the RF algorithm according to the measured values. In the accuracy metric, NB had the highest rate with 81%.

On the basis of classification algorithms, when percentage sampling and CV methods are compared;

(1) RF algorithm is the method with the highest performance in percentage and CV sampling.
(2) Accuracy, recall and F1 scores rates are higher in CV than the percentage algorithm in all classification algorithms.
(3) While the precision metric is higher in percentage sampling in LR and RF, CV has a higher rate in DT and NB.

When Tables 4 and 5 were examined, an increase in the performance of all classification algorithms was observed with data splitting with CV. A notable increase in accuracy was seen, especially in the DT and NB methods.

**Observation 2:** Classification with PER and 10-fold CV with PCA

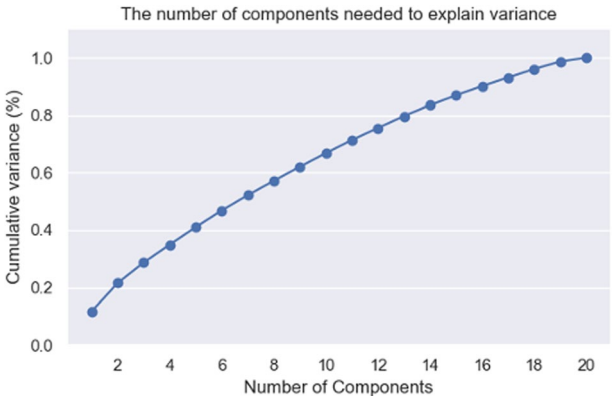


**Fig. 2** Number of components and variance for PCA

Figure 2 shows the number of principal components (PC) and variance relationship for PCA. The acceptable number of PCs and variance varies depending on the data set. This study focused on PCs with cumulative variance of 80% and above (Farquad et al. 2011). Starting from 13 PCs with a cumulative variance of 80%, up to 20 PCs were examined, respectively.

According to Tables 6 and 7, which include the results of using PCA with percentage sampling and CV, it was seen that the use of PCA was sufficient for all classification algorithms with 13 PCs with 80% cumulative variance. When using PCA with percentage sampling, the highest accuracy was measured in RF with 74%. In other metrics, RF also outperformed other classification algorithms.

When using PCA with CV, the best accuracy was calculated in the NB algorithm with 74%, and the second highest rate was calculated in the RF algorithm with 73%. In other parameters, all algorithms showed a similar performance.

Figure 3a shows the ROC curves of the classification results with PER, Fig. 3b with CV, Fig. 3c with PCA and PER, and Fig. 3d with PCA and CV. AUC ratios obtained from ROC curves are also shown.

According to Tables 6 and 7, when LR, RF and NB classification algorithms were used, it was seen that the effect of using 13 PCs, obtained with 80% of the cumulative variance, on performance gave results close to or better than the results obtained with raw data.

The summary chart of all classification algorithms, PER and CV, and tests performed with and without PCA is shown in Fig. 4.

## 5 Discussion

With the rapid growth and development of the credit industry, the performance of credit scoring models has gained great importance for financial profitability. Credit scoring has become an increasingly important tool for distinguishing between good and bad applicants and therefore managing credit risk. In this study, machine learning methods used in credit risk assessment were investigated. Then, inferences were made by applying various methods to the German data set.

**Table 6** Results obtained with PCA and PER based on the number of PCs

| | DT | | | | | | | | LR | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PC size | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| ACC (%) | 0.65 | 0.64 | 0.63 | 0.65 | 0.67 | 0.64 | 0.66 | 0.65 | 0.73 | 0.73 | 0.73 | 0.71 | 0.72 | 0.69 | 0.70 | 0.70 |
| PREC (%) | 0.71 | 0.72 | 0.71 | 0.74 | 0.74 | 0.73 | 0.74 | 0.72 | 0.93 | 0.93 | 0.92 | 0.92 | 0.92 | 0.90 | 0.90 | 0.89 |
| REC (%) | 0.77 | 0.76 | 0.74 | 0.76 | 0.77 | 0.75 | 0.76 | 0.76 | 0.74 | 0.75 | 0.75 | 0.73 | 0.74 | 0.73 | 0.73 | 0.73 |
| F1 (%) | 0.74 | 0.74 | 0.73 | 0.75 | 0.76 | 0.74 | 0.75 | 0.74 | 0.83 | 0.83 | 0.83 | 0.82 | 0.82 | 0.80 | 0.80 | 0.80 |
| | RF | | | | | | | | NB | | | | | | | |
| PC size | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| ACC (%) | 0.74 | 0.73 | 0.73 | 0.73 | 0.72 | 0.74 | 0.74 | 0.74 | 0.65 | 0.64 | 0.63 | 0.66 | 0.67 | 0.64 | 0.66 | 0.65 |
| PREC (%) | 0.94 | 0.92 | 0.94 | 0.93 | 0.94 | 0.94 | 0.94 | 0.95 | 0.71 | 0.72 | 0.71 | 0.74 | 0.74 | 0.73 | 0.74 | 0.72 |
| REC (%) | 0.75 | 0.75 | 0.74 | 0.74 | 0.73 | 0.75 | 0.75 | 0.75 | 0.77 | 0.76 | 0.74 | 0.76 | 0.78 | 0.75 | 0.76 | 0.76 |
| F1 (%) | 0.83 | 0.83 | 0.83 | 0.83 | 0.82 | 0.83 | 0.83 | 0.84 | 0.74 | 0.74 | 0.73 | 0.75 | 0.76 | 0.74 | 0.75 | 0.74 |

**Table 7** Results obtained with 10-fold CV and PCA based on number of PCs

| | DT | | | | | | | | LR | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PC size | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| ACC (%) | 0.66 | 0.67 | 0.66 | 0.66 | 0.67 | 0.66 | 0.66 | 0.66 | 0.72 | 0.72 | 0.72 | 0.72 | 0.72 | 0.72 | 0.72 | 0.71 |
| PREC (%) | 0.75 | 0.76 | 0.77 | 0.77 | 0.76 | 0.76 | 0.76 | 0.77 | 0.75 | 0.75 | 0.75 | 0.75 | 0.75 | 0.75 | 0.75 | 0.75 |
| REC (%) | 0.77 | 0.76 | 0.75 | 0.75 | 0.77 | 0.75 | 0.75 | 0.75 | 0.90 | 0.90 | 0.90 | 0.90 | 0.90 | 0.89 | 0.89 | 0.88 |
| F1 (%) | 0.76 | 0.76 | 0.76 | 0.75 | 0.76 | 0.76 | 0.75 | 0.76 | 0.82 | 0.82 | 0.82 | 0.82 | 0.82 | 0.82 | 0.81 | 0.81 |
| | RF | | | | | | | | NB | | | | | | | |
| PC size | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| ACC (%) | 0.73 | 0.71 | 0.74 | 0.74 | 0.71 | 0.72 | 0.73 | 0.73 | 0.74 | 0.73 | 0.73 | 0.74 | 0.73 | 0.73 | 0.74 | 0.72 |
| PREC (%) | 0.75 | 0.74 | 0.76 | 0.76 | 0.75 | 0.75 | 0.75 | 0.75 | 0.76 | 0.76 | 0.76 | 0.76 | 0.76 | 0.77 | 0.77 | 0.78 |
| REC (%) | 0.91 | 0.90 | 0.93 | 0.90 | 0.90 | 0.91 | 0.92 | 0.90 | 0.91 | 0.91 | 0.90 | 0.91 | 0.90 | 0.89 | 0.89 | 0.83 |
| F1 (%) | 0.82 | 0.81 | 0.83 | 0.83 | 0.81 | 0.82 | 0.83 | 0.82 | 0.83 | 0.83 | 0.82 | 0.83 | 0.83 | 0.82 | 0.82 | 0.81 |

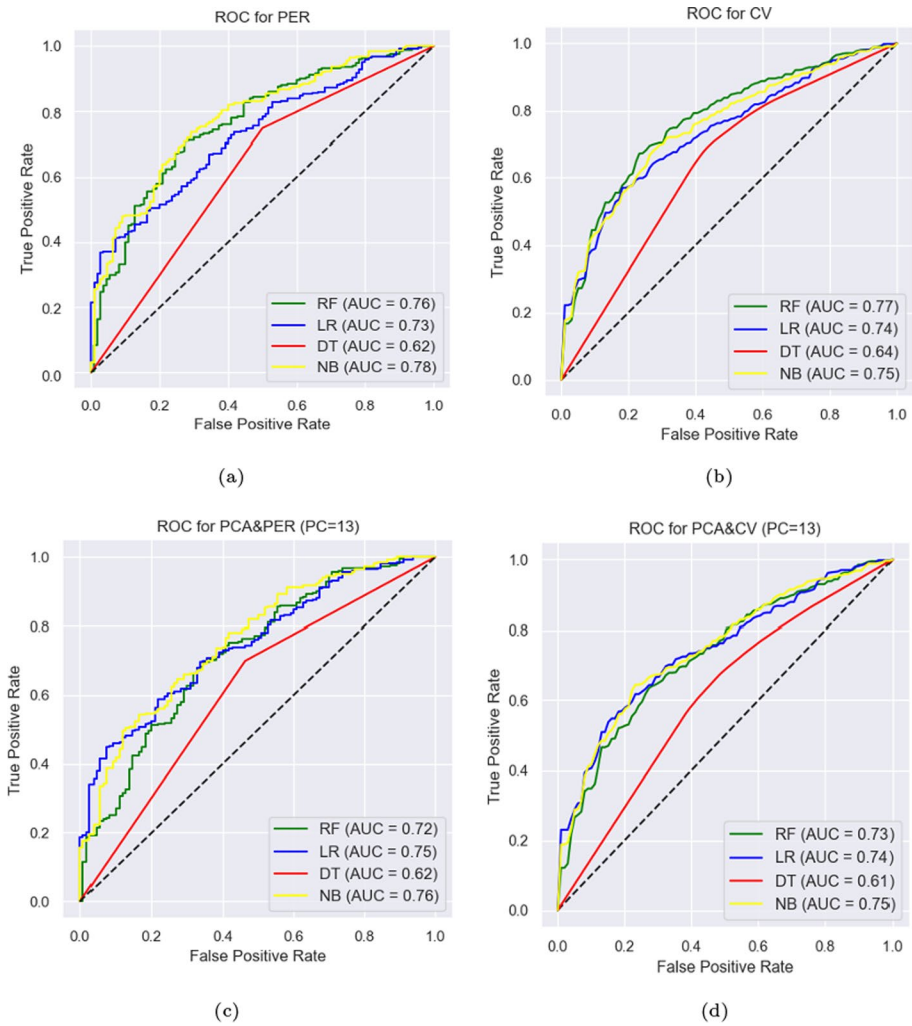


**Fig. 3** **a** ROC for PER, **b** ROC for CV, **c** ROC for PCA and PER, and **d** ROC for PC and CV

The study investigated the effect of dimensionality reduction by data splitting on the classification performance in credit risk assessment. PER and CV were chosen as the data splitting method and PCA was chosen as the dimensionality reduction method. The results for the cases detailed in the "Results" section are summarized in Tables 8 and 9 on the basis of accuracy and AUC.

In PER, CV, PCA and PER cases, the highest accuracies of 74%, 75%, and 74% were obtained with the RF algorithm. In the case of PCA and CV, the highest accuracies were observed in NB with 74% and RF with 73%. Based on these results, it can be concluded that the RF algorithm can be a successful choice for CRA in all cases (Appel et al. 2019; Ampountolas et al. 2021; Moscato et al. 2021; Teles et al. 2021).

It was observed that the use of CV and PCA in the NB algorithm provides an increase in accuracies. The highest accuracy obtained with the NB algorithm was calculated as

**Fig. 4** Results obtained in all cases

**Table 8** Summary of accuracy rates

| | DT | LR | RF | NB |
|---|---|---|---|---|
| PER | 0.67 | 0.70 | 0.74 | 0.67 |
| CV | 0.71 | 0.71 | 0.75 | 0.71 |
| PCA[a] and PER | 0.65 | 0.73 | 0.74 | 0.65 |
| PCA[a] and CV | 0.66 | 0.72 | 0.73 | 0.74 |

[a]For 13 PCs

**Table 9** Summary of AUC rates

| | DT | LR | RF | NB |
|---|---|---|---|---|
| PER | 0.62 | 0.73 | 0.76 | 0.78 |
| CV | 0.64 | 0.74 | 0.77 | 0.75 |
| PCA[a] and PER | 0.62 | 0.75 | 0.72 | 0.76 |
| PCA[a] and CV | 0.61 | 0.74 | 0.73 | 0.75 |

[a] For 13 PCs

74% with PCA and CV. It was shown that the use of CV and PCA may be appropriate to increase the performance of the NB algorithm (Yan et al. 2013).

It was determined that CV gives better accuracy results than PER in all classification algorithms.

Our study has shown that dimensionality reduction with PCA can be used especially in high-dimensional CRA datasets. Considering the dataset used, it has been concluded that PCs with a cumulative variance of 80% and above can represent the entire dataset.

When we look at the classification algorithms, it is seen that RF algorithm stand out in accuracy and NB algorithm stand out in AUC.

In the study, PER and CV methods have yielded similar results.

Also, it has been shown that dimensionality reduction can be effective in CRA studies and can be a stimulating perspective for future studies.

## 6 Conclusion

The purpose of the study was to investigate the effect of data splitting and dimension reduction operations on classification performance in credit risk assessment. German credit data set was used in the study.

As the data splitting method, 10-fold cross validation and percentage sampling were selected. PCA was used for dimension reduction. Also, classification was done with DT, LR, RF and NB algorithms.

The study consisted of four stages: (1) Classification by PER, (2) Classification by CV, (3) Classification by PCA and PER, and (4) Classification by PCA and CV. Classification results at each stage were evaluated in terms of accuracy, precision, recall, F1 score, and AUC metrics.

RF algorithm was the method with the highest performance in percentage and CV sampling. Accuracy, recall and F1 scores rates were higher in CV than the percentage algorithm in all classification algorithms. While the precision metric was higher in percentage sampling in LR and RF, CV has a higher rate in DT and NB.

It was observed that dimension reduction with PCA had a positive contribution to credit risk assessment. The data set consisting of 20 features was expressed by 13 PCs, accounting for 80% of the cumulative variance. In the NB algorithm, it was observed that the use of PCA and CV increases the accuracy rate (74%). With the use of PCA and PER in the LR algorithm, the accuracy rate increased and was calculated as 73%. In the RF algorithm, similar results were obtained at all stages. Therefore, it was concluded that PCA could be a preferable method for dimension reduction in the RF algorithm.

When examined on the basis of classification algorithms,

(1) In the DT algorithm, high accuracy of 71% was achieved with CV. An accuracy of 67% was observed with PER, 65% with PCA and PER, and 66% with PCA and CV. With the DT algorithm, it was concluded that the best choice was CV.
(2) Similar accuracy rates were obtained in the tests performed using the LR algorithm. The highest accuracy was measured as 73% with PCA and PER. 70% accuracy was achieved with PER, 71% with CV and 72% with PCA and CV.

(3) When the tests in the RF algorithm were examined, it was seen that similar results were obtained. The highest accuracy was 75% with CV, followed by 74% with PER and PCA and PER, and 73% with PCA and CV.
(4) In the NB classification algorithm, when PCA and CV was used, a significantly increased accuracy (74%) was achieved compared to the original dataset. The second highest rate was 71% with CV. An accuracy of 67% was observed with PER and 65% with PCA and PER.

In this study, it was concluded that the use of PCA can be preferred for LR, RF and NB algorithms. It has been observed that especially the use of PCA and CV in the NB algorithm gives higher accuracy than the original dataset. It was observed that the CV method was effective in the DT classification algorithm. One of the most important results of this study is that the cumulative variance was selected as 80% when reducing the dimension with the PCA method. Thus, positive results were obtained with less variance and PC.

The study is significant because of the very promising outcomes achieved. The future studies can be carried out on the evaluation of different classification algorithms and dimension reduction and data splitting methods in credit risk assessment, testing the same stages with different data sets, and the optimum number of PCs for PCA.

**Author contributions** Authors contributed equally.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflict of interest** This article was prepared with the support of 100/2000 YÖK Doctoral Scholarship.

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